Fall 2023 CSE340 Online Midterm Exam.

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Sec: 08

Set: A

Ans to the Q.No-1

1. (a) Elapsed time is the total time of a program's execution time, OS time, I/O time, idle time and waiting time.

CPU time is only the execution time of a program.

1. (b) Power tend, $P = \text{Capacitive load} \times \text{Voltage}^2 \times \text{frequency}$

Given,

Here,

$$\frac{P_{new}}{P_{old}} = \frac{0.88 \times C_{old} \times 0.79 f_{old}}{}$$

Here,

$$\frac{P_{new}}{P_{old}} = \frac{(0.88 \times C_{old}) \times (0.79 \times V_{old})^2 \times (0.79 \times f_{old})}{C_{old} \times V_{old}^2 \times f_{old}}$$

$$= 0.43$$

$\therefore$ Reduced Power $= (1 - 0.43) \times 100\%$

$= 57\%$ (Ans.)

1. (c) Given,

$T_{total} = 120 \text{ sec}$

$T_{unaffected} = (120 \times 17\%)$

$= 20.4 \text{ sec}$

$\therefore T_{improve} = \frac{120}{2.56}$

$= 46.875 \text{ sec}$

$\therefore T_{affected} = (120 - 20.4)$

$= 99.6 \text{ sec}$

Now,

$$T_{improve} = \frac{T_{aff}}{n} + T_{unaff}$$

$$\Rightarrow 46.875 = \frac{99.6}{n} + 20.4$$

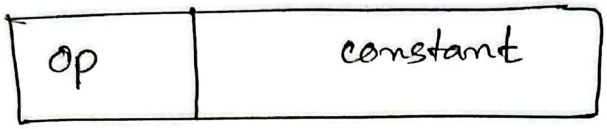
$$\Rightarrow n = 3.762$$

$\therefore$ I have to improve 3.762 times.

Ans.

Ans to the Q.No-12

2. (a) Format of J-type,



In MIPS, for Jump address the target address is calculated by taking the MSB 4 bits from the PC and add the 4 bit in the MSB of that constant address of 28 bit and make it 32 bit. But before 28 bit it shift 2 bit left of the 26 bit constant address. Then the 32 bit address is added with the PC address and we get the target address.

Given, PC = 0x00000000

~~JAL 100~~

JAL 120

the constant part = $(120)_d$

in 26 bit = $(0000 \cdots 0111\ 1000)_b$

after doing 2 bit left shift, in 28 bit,

$(0000\ 0000\ 0000\ 0000\ 0001\ 1110\ 0000)_b$

adding the MSB 4 bits from PC to the MSB of 120,

in 32 bit $= (0000\ 0000\ 0000\ 0000\ 0000\ 0001\ 1110\ 0000)_b$

in Hex $= (0x000001E0)$

Now,

target address $= (0x000001E0 + 0x00000000)$

$= (0x000001E0)$

(Ans.)

2. (b) Given, $6 = 12
$9 = 4
$5 = ?

16
128
32

```
add $5, $6, $9   → $5 = 16
sll $5, $5, 3    → $5 = 128
srl $5, $5, 2    → $ = 32
lw  $5, 40($6)   → $5 = 32 and memory address = (0x000001E0)
```

2.(c) Given,

$x = \underbrace{1234}_{H}\underbrace{5678}_{L}{}_{H}$ (in $4)

MIPS:

```
lui $4, 0x1234
ori $4, $4, 0x5678
```

2.(d) Given,

$A[3] = 12X - 7A[5] + 33Y - 70$

A → $s0
X → $s1
Y → $s2

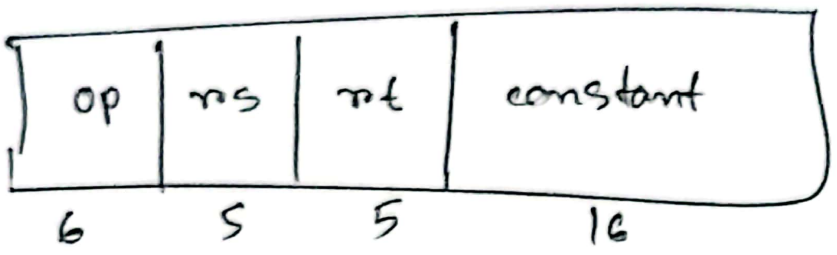
MIPS:

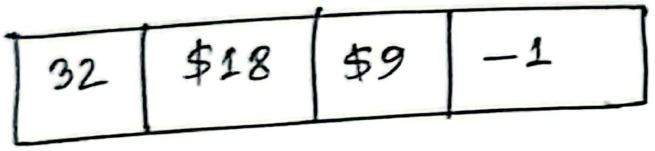
```
sll $t0, $s1, 3
sll $t1, $s1, 2
add $t1, $t0, $t1      → t1 = 13X
lw $t2, 20($s0)        → t2 = A[5]
sll $t3, $t2, 3        → t3 = 8A[5]
sub $t3, $t3, $t2      → t3 = 7A[5]
sll $t4, $s2, 5        → 32Y = t4
add $t4, $t4, $s2      → t4 = 33Y
sub $t1, $t1, $t3
add $t1, $t1, $t4
addi $t1, $t1, -70
sw $t1, 12($s0)        → A[3] = ...
```

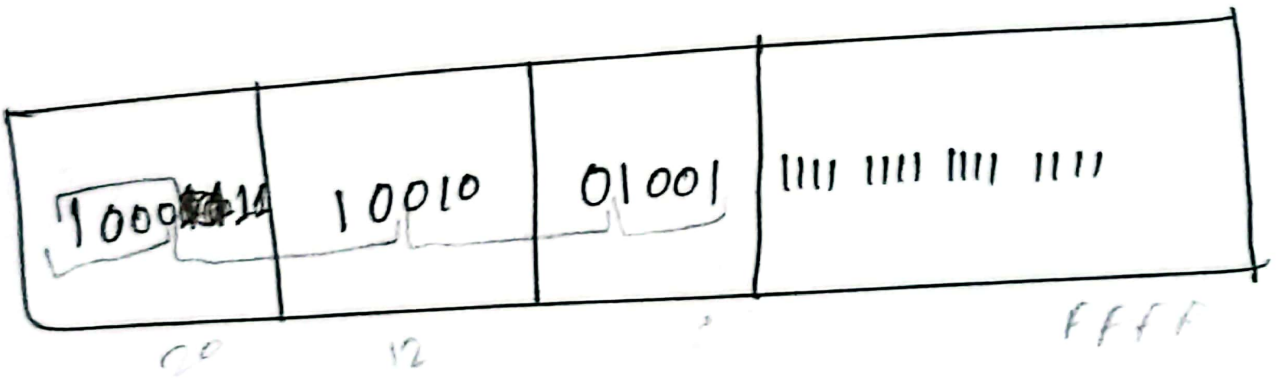
2. (c) Given,

addi $9, $18, -1 (op=32)

this is an i-type instruction.







Now, 1000 1110 0100 1001 1111 1111 1111 1111

in Hex = 0x8E49FFFF

Ans.